SIMATSENGINEERING

ASSESSMENTTOOL1:AnalyticalProblemSolving

COURSENAME: CloudComputingand BigDataAnalytics **COURSECODE:** CSA15

COURSEOUTCOMECOVERED

CO2:*Analyzevirtualizationarchitecturestoidentifyperformanceandsecurityissuesincloudcomputing environments.* (BL3)

Dave'sTaxonomy:Manipulation(Level2) **TotalMarks:50**
QuestionCount:5
Duration:60Minutes

CodeofConduct

I _____(Name/RegNo)certifythatthissubmissionismyoriginalworkandthatIhaveadhered to the guidelines specified for this assessment.
Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.
Signature of the student: _____

Assessment Tasks

AnalyticalProblemSolving

1. CompareType-1andType-2hypervisorsforaclouddatacenterhostingmission-criticalapplications.Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware.Analyzehowvirtualizationarchitecturesupportsisolationandidentifytwomajorattacksurfaces.
3. StudythefollowingVMhostutilizationsnapshotandrecommendcorrectiveactions:CPU92%,memory88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. AnalyzehowSoftwareDefinedDatacenterarchitectureimprovesagilitycomparedwithatraditionaldatacenter. Support your answer with compute, storage, and network virtualization considerations.
5. Amulti-cloudfederationprojectfailsduetoidentitymismatchesbetweenproviders.Analyzetheissueand propose a secure identity and privacy design using federation concepts.

EquivalentStepsinVMwareWorkstationPro

- LibrarypaneshowingmultipleVMs=multipleinstances/regions.
- VM→Snapshot→TakeSnapshot,thenSnapshotManager=backup/versioningdemonstration.
- Edit→VirtualNetworkEditortoshowNAT/Host-only/Bridgednetworks=hybridormulti-cloud connectivity.
- VM→Manage→Clone=replica/auto-scalingdemonstration.

Tip:Pasteeachscreenshotdirectlyundertherelevantquestioninthisdocument(Insert→PictureinWord)before final submission, with a one-line caption explaining what it demonstrates.

AnalyticalProblemSolvingRubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interpretsall technical requirements accurately	Interpretsmost requirements correctly	Partial interpretation	Misinterpretsthe problem
Analytical Reasoning	25%	Strongreasoning with clear cause-effect analysis	Goodreasoning withminorgaps	Basicanalysis only	Weakorno analysis
MethodSelection	20%	Chooses highly appropriate virtualizationand securityconcepts	Chooses mostly suitableconcepts	Partiallysuitable concepts	Inappropriate approach
SolutionQuality	20%	Providesfeasible and technically sound solution	Reasonable solutionwith minor issues	Basic solution withlimitations	Unclearorinvalid solution
Presentationof Analysis	15%	Well-structured andtechnically precise	Generallyclear	Somewhatunclear	Difficulttofollow

TotalMarksAwarded:

Answers

Question1:Type-1vsType-2Hypervisors(10Marks)

Probleminterpretation

Adatacenterhostingmission-criticalapplicationsneedshighperformance,strongisolation,high availability and centralized control. The hypervisor type directly affects all of these.

Comparison

Criteria	Type-1(Bare-metal)	Type-2(Hosted)
Examples	VMwareESXi,MicrosoftHyper-V,KVM,Xen	VMwareWorkstation,VirtualBox,Parallels
Architecture	Runsdirectlyonhardware;VMssitonthe hypervisor	RunsasanapplicationonahostOS;VMssit on hypervisor + host OS
Performance	Near-native;directaccesstoCPU,memory, I/O; no host OS overhead	Extra layer (host OS) adds latency and overhead;hostOSprocessescompetefor resources
Security	Small attack surface (thin, purpose-built kernel);hardware-assistedisolation(Intel VT-x/AMD-V)	Largeattacksurface:avulnerabilityinthe hostOSoritsappscancompromiseevery VM
Manageability	Centralizedmanagement(vCenter,SCCM, OpenStack), live migration, HA, DRS, clustering,automation	Manual,per-machinemanagement;no clustering or live migration at scale
Reliability	Fewercomponents,sofewerfailurepoints; supports HA failover	HostOScrash,patchorreboottakesdown all VMs
Typicaluse	Productiondatacenters,cloudproviders	Development,testing,training,desktops

Justification

Type-1isthebetterchoice. Mission-criticalworkloadsneedpredictableperformance,minimal downtime and strong isolation. Type-1 removes the host OS layer (less overhead and fewer vulnerabilities), supports live migration and high availability, and scales through centralized orchestration. Type-2 is suitable only for labs and development.

Question2:WorkloadIsolationandAttackSurfaces(10Marks)

Probleminterpretation

Finance,healthcareandstudentportalshavedifferentcomplianceneeds(e.g.,PCI-DSS,HIPAA,FERPA) but share the same physical hardware. The virtualization layer must enforce strict separation so one tenant cannot see or affect another.

Howvirtualizationarchitectureprovidesisolation

- **Computeisolation:** Eachtenantruns initsownVM withaseparateguestOS,so acompromisein one OS does not directly affect another.

- **Memory isolation:** The hypervisor uses hardware-assisted virtualization (Intel VT-x/AMD-V, EPT/NPT)soeachVMseesonlyitsownaddressspace; memoryballooningandpagesharingare controlled and monitored.
- **Storageisolation:** Separatevirtualdisks(VMDKs/VHDs)withper-tenantdatastores,LUNmasking, and encryption at rest.
- **Networkisolation:**Separatevirtualswitches,VLANs/VXLANsegments,virtualfirewallsandmicro-segmentation per tenant.
- **Resourceandadministrativeisolation:** Role-basedaccesscontrol,per-tenantresourcepools, quotas, and reservations prevent a noisy neighbor from exhausting resources.
- **Placement policy:** Critical workloads can be placed on dedicated hosts or clusters, with affinity/anti-affinityrules,forthestrongestseparation (e.g.,healthcareonitsown cluster).

Twomajorattacksurfaces

1. Thehypervisoritself(VM escape/hypervisorcompromise). Avulnerabilityinthehypervisororits device emulation (virtual NIC, graphics, USB) could allow malicious code in one VM to break out and access the host or other tenants' VMs. This would breach finance and healthcare data. **Mitigation:** timely patching, minimal hypervisor footprint, disabling unused virtual devices, secure boot, and hardened management.

2. Shared resources and virtual network (side-channel and VM-to-VM attacks). Shared CPU caches, memoryandvirtualswitchescanleakinformation(cachetimingattackssuchasSpectre/Meltdown-class issues), and weak virtual-network segmentation allows lateral movement or sniffing between VMs. **Mitigation:** VLAN/VXLAN segmentation, micro-segmentation firewalls, dedicated hosts for sensitive tenants, CPU microcode updates, and disabling cross-VM features such as page sharing where not needed.

Othersurfacesworthnoting:managementinterfaces/APIs,VMsnapshotsandimages,andVM migration traffic (should be encrypted).

Question3:VMHostUtilizationAnalysis(10Marks)

Observedsnapshot

Metric	Value	Interpretation
CPUutilization	92%	Criticallyhigh;littleheadroom;likelyCPU contention (high ready time)
Memoryutilization	88%	High;riskofballooning,swappingor compression
StorageI/Olatency	35ms	Poor;healthyrangeisroughlyunder10to20ms
AverageVMboottime	170s	Veryslow;typicalwell-sizedhostsbootinwell under a minute

Likelybottlenecks(cause-effectanalysis)

- **CPUcontention:**Thehostisovercommitted(toomany vCPUsperphysicalcore).VMswaitinthe scheduler queue, so applications slow down.

- **Memory pressure feeding storage load:** At 88%, the hypervisor starts reclaiming memory by ballooning,pagesharing,compressionandswappingtodisk.Swappingpushesadditional I/Oonto the already slow storage.
- **StorageI/Obottleneck:** 35mslatencysuggestsasaturatedorpoorlyconfigureddatastore(too many VMs per LUN, slow disks, no caching, or a congested SAN/network path).
- **Slow boot as the combined symptom:** Booting needs many random reads (OS image, services). Withhighlatency,CPUcontentionandmemoryswappingallatonce,boottimeinflatesto170s. This is the visible symptom of the three underlying problems; a boot storm can worsen it.

Primarybottleneck:storageI/OcombinedwithCPU/memoryovercommitment.Theseresources compound each other.

Correctiveactions

Area	Recommendedaction
CPU	Load-balance with DRS/live migration to less loaded hosts; right-size VMs (reduceunnecessaryvCPUs);lowerthevCPU:pCPUratio;addhostsorscaleout
Memory	Right-sizeVMmemory;eliminateidleVMs;addRAMorhosts;setreservations for critical VMs; avoid overcommitment beyond safe limits
Storage	Move to SSD/NVMe or tiered storage; spread VMs across datastores (Storage DRS);use caching;fixSAN/multipath/networkcongestion;use thin/linkedclones appropriately
Boottime	StaggerVMstart-up;usetemplate/linkedclonesandfaststorage;remove unnecessary startup services
Monitoring	Track CPUready,ballooning/swap,datastorelatency,setalertsat~75to80%, and capacity-plan ahead of growth

Question4:SoftwareDefinedDatacentervsTraditionalDatacenter(10Marks)

Probleminterpretation

Agilitymeanshowquicklyinfrastructurecanbe provisioned,changed,scaledandrecovered.ASoftware Defined Datacenter (SDDC) virtualizes and abstracts compute, storage and networking, and controls them through software, policy and automation.

Comparison

Aspect	TraditionalDatacenter	SoftwareDefinedDatacenter
Provisioning	Manual,hardware-driven;daystoweeks	Automated,template/API-driven;minutes
Compute	Physicalservers,lowutilization(about10to 20%), one app per server	Servervirtualization,consolidation,live migration, auto-scaling
Storage	SiloedSAN/NASmanagedperarray	Software-definedstorage(vSAN,Ceph); policy-basedpooling,tiering,snapshots
Network	Hardwareswitchesconfigureddeviceby device; VLAN changes are slow	Networkvirtualization(VXLAN,NSX/SDN); central controller, programmable, micro-segmentation

Aspect	TraditionalDatacenter	SoftwareDefinedDatacenter
Management	Manytools,humanconfiguration,error-prone	Unifiedorchestrationandpolicy-based automation
Scaling&DR	Buyandrackhardware;slowrecovery	Elasticscale,cloning,snapshots,replication, fast failover
Cost	HighCapEx,vendorlock-in	Commodityhardware,lowerOpEx,better utilization

Agilitygainsbyvirtualizationlayer

- **Computevirtualization:** VMs/containers are createdfrom templates or clones inminutes; resourcesarepooledandreallocateddynamically;DRSbalancesloadandvMotionenables maintenance without downtime.
- **Storagevirtualization:** Storageisabstractedintopools andprovisionedbypolicy(performance, availability); thin provisioning, snapshots and replication give fast backup and recovery.
- **Network virtualization:** Virtual networks, firewalls and load balancers are created in software withoutre-cabling;securitypoliciesfollowtheVM,whichmakesmulti-tenantisolationandhybrid connectivity faster.

Conclusion: SDDCimprovesagilitythroughabstraction,automationand centralizedpolicycontrol,giving faster provisioning, elastic scaling, easier recovery and better utilization than a hardware-bound traditional datacenter.

Question5:Multi-CloudFederationIdentityMismatch(10Marks)

Problemanalysis

Each cloud provider (e.g., AWS, Azure, GCP) has its own identity system, user formats, role models, attribute names, and token formats. In a federation project, the same user may have different identifiersineachprovider(e.g.,email vsUPNvsuniqueID),differentroles/permissionsemantics,and incompatible authentication protocols. This causes failed authentication, wrong or excessive access, duplicate accounts, broken trust, and compliance exposure when personal data is shared without control.

Rootcauses

- Nocommonidentitystandardortrustrelationshipbetweenproviders
- Inconsistentattributeandrolemapping(nocanonicalschema)
- Localcredentialsstoredseparatelyineachcloud(identitysilos)
- Mismatchedprotocols/tokenformats(SAMLvsOAuth/OIDC),clockskew,certificateormetadata errors
- Unmanagedlifecycle(de-provisioningnotsynchronized)andexcessiveattributesharing

Proposedsecureidentityandprivacydesign

- **1.Federatedtrustmodel:** UseacentralIdentityProvider(IdP)suchasAzureAD/Okta/Keycloakas the single source of truth. Each cloud acts as a Service Provider (SP) that trusts the IdP.
- **2.Standardprotocols:** UseSAML2.0/OpenIDConnect/OAuth 2.0forauthenticationand authorization, with signed (and encrypted) assertions/tokens and short lifetimes.

- **3. Identity mapping and synchronization:** Use SCIM for automated provisioning and de-provisioning,anddefineacommonattribute schema withamappingtable(e.g.,canonicaluserID to each provider's format).
- **4.Roleandaccessmapping:** Mapfederatedgroupsto least-privilegerolesineachcloud (RBAC/ABAC); use cross-account role assumption instead of shared credentials.
- **5.Strongauthentication:** EnforceMFA,conditionalaccess,andSingleSign-On(SSO);use certificate/metadata validation and NTP time sync.
- **6. Privacy protection:** Release only the minimum required attributes (data minimization), use pseudonymousorpairwiseidentifiers,userconsent,encryptionintransit(TLS)andatrest,and data-residency controls to meet GDPR/HIPAA.
- **7.Auditingandgovernance:** Centralizedlogging,SIEM,regularaccessreviews,keyrotationand revocation lists.

Expectedresult: Usersauthenticateoncethrough the trustedIdPandreceiveconsistent,least-privilege access in every provider, with synchronized lifecycle management, auditability and protected personal data.

ScreenshotGuide(VMwareWorkstationPro)

VMwareStep	Demonstrates
LibrarypanewithmultipleVMs	Multipleinstances/regions
VM→Snapshot→TakeSnapshot,then Snapshot Manager	Backupandversioning
Edit→VirtualNetworkEditor(NAT /Host-only / Bridged)	Hybridormulti-cloudconnectivityandisolation
VM→Manage→Clone	Replica/auto-scaling